Milestone 7: Review Findings and Actions

What the reviews found, what we changed, and what remains open

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RESEARCH FRAMEWORK. Lift is research-generated or taken from published trials; no InComm lift, financial or issuance data is used. Dollar values are research ranges tagged by provenance.

1. Where we are

- Measured lift on the research mock: ATE 0.0114 (SE 0.0187); minimum detectable 0.0524. Status: LIFT_NOT_DISTINGUISHABLE_FROM_ZERO. No measured dollar ROI is claimed.
- Star value of one closure: about $157 at base ($102 to $260), measured on CMS public 2026 data. Break-even lift is about 5.5 percentage points offered to all (3.3 to 8.4).
- Under published incentive-trial effects (CRC-anchored RR 1.17) the program nets $-24,733 at base value (ROI -0.22x). It pays for DIAB_A1C_TEST, DIAB_BLOOD_SUGAR and HTN_PCP_FOLLOWUP and not for the other gaps.
- Method: every input comes from the reference tables in the InComm database, public CMS/KFF/HCUP data, or published trials, and is labeled by source.

2a. InComm feedback on Milestone 7 and our response

Point	Finding	Action
Eligible-share assumption drives the $157 per closure	Correct. The share was a fixed public anchor (diabetes 26%), and a plan with an SNP-heavy mix can differ.	Plan-supplied share is now an input. At a diabetes share of 40% the mean value is about $114 per closure and the break-even lift is 1.38 times the base (about 7.6pp). Sweep published from 20% to 50%.
Value should be conditional on a contract's star position	Correct. The headline averages over all 524 contracts, but only 3.5-star contracts can cross the 4-star boundary, and most of those have the measure far from a cut point.	Value now published by position. A 3.5-star contract with the measure near a cut point is worth about six times the average; other positions get $0 from this route. Downside protection for 4.0-star contracts is not modelled and is stated.

Redemption rate 0.988 is a proxy for OTC activation	Correct. Activation is not reward redemption, and 0.55 and 0.75 are equally unsourced.	0.988 is now labeled the cost ceiling. The break-even lift is reported as a range (5.5pp at 0.988, 4.2pp at 0.75, 3.1pp at 0.55). Redemption is an input.
Zero claims anomalies is circular on mock data	Correct. Closure is derived from the same claim, CPT and date the checks read back, so member, CPT, date and window agree by construction.	Renamed a referential-integrity check, not a validation. A fault-injection test shows the checker flags known faults (100% of six fault types, 0% false positives).

2. InComm feedback (28 September 2026) and our response

Point	Finding	Action
Invented simulation parameters; rolling period is at least a year	Correct: the 90-day window was an error. FundLifeSpan in InComm's wallet configuration is 365 days.	Window is 365 days; 90 and 180 days are sensitivity only. Amounts drawn from InComm's own per-gap pools. Offer probability and lag are assumptions, swept over 9 combinations: null in all.
Elasticity reported NOT_AVAILABLE	Every offer used one amount per gap, so no dose-response existed.	Within-gap fixed-effects estimator on varying amounts; minimum detectable slope about 1.3pp per $10. Real response curve needs a randomized amount arm (design written).
Diabetes cohort had an empty untreated arm	An artifact: lifetime reward counts were copied onto every gap of a member.	Gap-level exposure; every diabetes gap has both arms. A positivity table and Manski bounds now handle any genuinely empty arm.
CKD excluded	ConditionCode CKD versus CHRONIC_KIDNEY_DISEASE mismatch means InComm's rules engine never produced a CKD row.	Run as a separate secondary cohort on research labels; never mixed into the headline.

3. Defects we found ourselves and fixed

Finding	Effect	Fix
Class balancing in the AIPW nuisance models	Inflated the treatment effect to +3.3pp on a zero-effect mock and produced 52 'Treat' members	Removed; recovery tests (null, known effect, known heterogeneity, known dose-response) fail with the bug and pass without. M6 documents regenerated.
In-sample top-fraction uplift sums	Winner's curse: probability of net benefit near 1.0	Monte Carlo centered on the population ATE with its standard error.
A1C and blood sugar closures valued twice	Overstated Stars value	Measure value split across gaps sharing a measure.
Relative risk applied to a 74% baseline	Probabilities above 1	Effect carried on the odds-ratio scale.
InComm StarsWeight differs from CMS 2026 weights; two gaps feed no Part C measure	Overvalued eye exam and kidney screening; BH and COPD valued at nonzero	CMS weights used; discrepancy table published; BH and COPD Stars value is exactly 0.
Unsourced Stars value per closure	Value was a typed assumption	Boundary density measured on 524 CMS contracts; engine reproduces 98% of published ratings within half a star.
Wrong (V24) CMS table once fetched	Would have misstated RAF factors	Only V28 factors found in CMS materials used (diabetes 0.166, CKD stage 4 0.514); others stay NOT_AVAILABLE.

4. Items InComm raised or implied that M7 now covers

- Reward attribution: impossible on real data (claims precede cases); handled by probability of necessity and a recommendation to record offer timestamps.
- RAF: separate arm. Most gaps are laboratory or imaging tests that are not risk-adjustment eligible, so their RAF value is zero; HTN follow-up is not available without verified factors.
- Priced ROI: provided as labeled scenarios under published trial effects with break-even relative risk per gap, plus a separate avoided-cost arm for BH follow-up.

5. What remains open and why

- Measured lift: needs randomized offers. Sizing: about 1,500 members per arm detects +5pp pooled (design document).
- Structural limits of the value model: uniform position inside a half-star band, categorical adjustment and reward factor not modelled, no downside protection for 4.0-star contracts, single-measure marginal view.
- Independent claims checks need diagnosis and claim-status fields beyond what the closure derivation guarantees; the checker flags a claim reused across members as one that does.
- Avoided cost: BH only and observational; no verified link for other gaps. Not added to the headline.
- Synthetic data: claim IDs are MOCK-CLM and the IncentiveRewards ledger is research-generated, so clean validation describes the generator, not InComm production quality.
- Not confirmed: whether withholding offers from a control group is permitted by program rules.